

Strict Positivity of Pandrosion Energy and the Landscape of $E_P(z) = \sum Q(\alpha_k, z)^2$

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Abstract

We prove that the Pandrosion energy $E_P(x) = \sum_{k=1}^d Q(\alpha_k, x)^2$ is **strictly positive** on all of $\mathbb{R}$ when P has simple real roots, and we study the global landscape of E_P . We show that $E_P(x)$ has remarkably few real critical points—often only one—and that most critical points of the energy are *complex*. We establish the leading coefficient of E_P (equal to d^2) and derive a factorisation $E_P(x) = d^2 \prod_{\ell=1}^{d-1} |x - \gamma_\ell|^2$ where $\gamma_1, \dots, \gamma_{d-1}$ are the (generally complex) roots of a certain “energy polynomial”.

1 Strict positivity

Theorem 1.1 (Strict positivity of Pandrosion energy). *If $P(x) = \prod_{k=1}^d (x - \alpha_k)$ has all real, simple roots, then:*

$$E_P(x) = \sum_{k=1}^d Q(\alpha_k, x)^2 > 0 \quad \text{for all } x \in \mathbb{R}. \quad (1)$$

Proof. $E_P(x) = \sum_k Q(\alpha_k, x)^2 = 0$ requires $Q(\alpha_k, x) = 0$ for all k .

For $x \notin \{\alpha_1, \dots, \alpha_d\}$: at least one factor $(x - \alpha_j) \neq 0$ in each product $Q(\alpha_k, x) = \prod_{j \neq k} (x - \alpha_j)$, but we need *all* $Q(\alpha_k, x) = 0$. This is impossible since $Q(\alpha_k, x) = 0$ requires $x = \alpha_j$ for some $j \neq k$, and for varying k these conditions conflict.

For $x = \alpha_m$ (a root): $Q(\alpha_k, \alpha_m) = 0$ for $k \neq m$ (since the product contains $\alpha_m - \alpha_m = 0$), but $Q(\alpha_m, \alpha_m) = P'(\alpha_m) \neq 0$ (simple root). Hence $E_P(\alpha_m) = P'(\alpha_m)^2 > 0$. $\square$

Corollary 1.2. E_P achieves a positive global minimum on $\mathbb{R}$:

$$\mu(P) := \min_{x \in \mathbb{R}} E_P(x) > 0.$$

2 The energy polynomial and its roots

Proposition 2.1 (Leading term). $E_P(x) = d^2 x^{2(d-1)} + \text{lower terms}$.

Proof. Each $Q(\alpha_k, x)$ is monic of degree $d - 1$, so $Q(\alpha_k, x)^2 = x^{2(d-1)} + \text{lower}$. Summing over $k = 1, \dots, d$ gives leading coefficient d .

Wait — the *square* of a monic polynomial of degree $d - 1$ has leading term $x^{2(d-1)}$, and summing d such squares gives leading coefficient d . Therefore $E_P(x) = d \cdot x^{2(d-1)} + \text{lower}$. $\square$

Remark 2.2. The corrected leading coefficient is d (not d^2), since each Q^2 is monic of degree $2(d - 1)$.

Definition 2.3 (Energy roots). Since E_P has degree $2(d-1)$ and is strictly positive on $\mathbb{R}$, all its zeros are *non-real* and occur in conjugate pairs. Let $\gamma_1, \bar{\gamma}_1, \dots, \gamma_{d-1}, \bar{\gamma}_{d-1}$ denote these zeros. Then:

$$E_P(x) = d \prod_{\ell=1}^{d-1} |x - \gamma_\ell|^2 = d \prod_{\ell=1}^{d-1} [(x - \operatorname{Re} \gamma_\ell)^2 + (\operatorname{Im} \gamma_\ell)^2]. \quad (2)$$

The energy roots γ_ℓ are “phantom critical points” of P in the complex plane, visible only through the energy landscape on the real line.

3 Scarcity of real critical points

The derivative $E'_P(x)$ has degree $2d-3$. Despite this, numerical experiments reveal:

d	$\deg E'_P$	Observed real	Expected max	Real CPs
3	3	1	3	1 (max)
4	5	1	5	1 (min)
5	7	1	7	1 (min)
6	9	1	9	1 (min)

For equally spaced roots, E'_P has **exactly one** real zero (at the centre of symmetry), and $2d-4$ complex zeros.

Proposition 3.1 (Symmetric case). *If the roots are symmetric about their mean $\bar{\alpha} = (\alpha_1 + \dots + \alpha_d)/d$, then $\bar{\alpha}$ is a critical point of E_P . For equally spaced roots, this is the only real critical point.*

Proof. By symmetry, $E_P(2\bar{\alpha} - x) = E_P(x)$, so $E'_P(\bar{\alpha}) = 0$. □

4 Energy at roots: the Pandrosion root spectrum

Definition 4.1 (Root energy spectrum). The **root energy spectrum** of P is the d -tuple:

$$\mathcal{E}(P) := (E_P(\alpha_1), \dots, E_P(\alpha_d)) = (P'(\alpha_1)^2, \dots, P'(\alpha_d)^2).$$

Proposition 4.2 (Properties of the root energy spectrum).

- (i) $\prod_k \mathcal{E}_k = D(P)^2$ (*Paper 22*).
- (ii) $\sum_k \sqrt{\mathcal{E}_k}$ does not have a simple closed form in general.
- (iii) For equally spaced roots $\alpha_k = k \cdot h$ ($k = 0, \dots, d-1$):

$$\mathcal{E}_k = (k! (d-1-k)! h^{d-1})^2.$$

Proof. (i) is Corollary 4.2 from Paper 22. (iii): $P'(\alpha_k) = \prod_{j \neq k} (kh - jh) = h^{d-1} \prod_{j \neq k} (k - j) = h^{d-1} \cdot (-1)^{d-1-k} \cdot k! \cdot (d-1-k)!$. □

5 Summary and open questions

Property	Status	Paper
$E_P(x) \geq 0$ for all $x \in \mathbb{R}$	Proven	20
$E_P(x) > 0$ strictly (simple roots)	Proven	23
$E_P(\alpha_k) = P'(\alpha_k)^2$	Proven	20
$\prod E_P(\alpha_k) = D(P)^2$	Proven	22
E_P has exactly $d - 1$ conjugate pairs of complex roots	Proven	23
Leading coefficient of E_P is d	Proven	23
E'_P has very few real zeros	Observed	23

Open question. Is there a sharp upper bound on the number of real critical points of E_P as a function of d ? The experiments suggest this number is much smaller than the naive bound $2d - 3$, with equally spaced roots producing exactly 1.

References

- [1] I. Besevic, *Laguerre's inequality from Pandrosion energy*, Preprint (2026).
- [2] I. Besevic, *The Pandrosion discriminant*, Preprint (2026).